

NEWTON SCHOOL OF TECHNOLOGY

Data Visualisation Analytics — Capstone 2

Decoding Beijing's Air Crisis

A Data-Driven Analysis of PM2.5 Pollution Patterns (2010–2014)

Sector	Environmental Analytics / Public Health
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Section	D
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1 Executive Summary

Project Core: This analysis identifies that Beijing’s PM2.5 levels exceeded WHO safe limits by 6–7× annually (2010–2014). We establish wind speed and seasonal coal heating as the primary drivers of hazardous air quality.

1.1 Problem

Fine particulate matter (PM2.5) represents the single greatest environmental health risk in urban China. During 2010–2014, Beijing’s air quality remained chronically hazardous without a structured, data-backed decision framework for pollution alerts.

1.2 Approach

We executed an end-to-end analytics pipeline on 43,824 hourly observations from the UCI Beijing PM2.5 dataset. This included automated ETL cleaning, 11 business-focused EDA visualisations, 7 non-parametric statistical tests, and a dual-view Tableau dashboard.

1.3 Key Insights

- **WHO Breach:** Annual means (89.9–101.6 $\mu\text{g}/\text{m}^3$) were 6–7× higher than the 15 $\mu\text{g}/\text{m}^3$ guideline.
- **Seasonality:** Winter is the most polluted season (109.9 $\mu\text{g}/\text{m}^3$) due to heating emissions.
- **Wind Factor:** Calm wind conditions (cv) resulted in the highest pollution (124.5 $\mu\text{g}/\text{m}^3$), while NW winds act as a natural purifier (69.8 $\mu\text{g}/\text{m}^3$).

1.4 Key Recommendations

- Launch winter-priority emission reduction programmes targeting coal heating.
- Implement “Clean Window” alerts triggered by North-Westerly wind forecasts.
- Deploy calm-wind emergency protocols (traffic/industrial curtailment) when stagnation is predicted.

2 Sector and Business Context

2.1 Sector Overview

This project sits at the intersection of environmental analytics and public health policy. Air quality management is critical for industrialising cities where growth has historically been prioritised over environmental protection. China’s NAAQS annual limit is 35 $\mu\text{g}/\text{m}^3$, still well above the WHO’s 15 $\mu\text{g}/\text{m}^3$ standard.

2.2 Decision-Maker / Stakeholder

- **Beijing Municipal Environmental Bureau:** To determine pollution alert timing and seasonal curtailment.

- **Urban Public Health Officials:** To schedule advisories for elderly and respiratory patients.
- **Traffic Management:** To trigger vehicle restrictions during low-dispersion windows.

2.3 Why This Problem Matters

Beijing’s PM2.5 frequently exceeded the “Very Unhealthy” threshold ($250\text{ }\mu\text{g}/\text{m}^3$), contributing significantly to cardiovascular and respiratory mortality in China. Without a data-backed alert framework, responses remain reactive.

3 Problem Statement and Objectives

3.1 Formal Problem Definition

Core Question: Which meteorological conditions and temporal patterns best predict hazardous PM2.5 levels in Beijing, and how can these be operationalised into a decision-support framework?

3.2 Scope

- **In scope:** Hourly PM2.5 data (US Embassy, Beijing); meteorological variables (2010–2014); Tableau dashboard construction.
- **Out of scope:** Multi-station spatial interpolation; real-time streaming; predictive ML modelling; economic cost modelling.

3.3 Success Criteria

Criterion	Target	Achieved	Status
Dataset Scale	>5,000 rows	43,824 rows	✓
KPIs Computed	≥ 4 KPIs	6 KPIs	✓
Statistical Tests	≥ 3 methods	7 analyses	✓
Decision Insights	8–12 insights	10 insights	✓
Actionable Recs	3–5	5 recs	✓

4 Data Description

4.1 Source and Access

- **Source:** UCI Machine Learning Repository (Liang et al., 2015).
- **Access URL:** <https://archive.ics.uci.edu/ml/datasets/Beijing+PM2.5+Data>
- **Granularity:** Hourly readings from the US Embassy, Beijing.

4.2 Dataset Size and Coverage

- **Dimensions:** 43,824 rows $\times$ 13 columns.
- **Temporal Range:** 1 January 2010 to 31 December 2014.

4.3 Key Columns

Column	Description	Role
pm2.5	PM2.5 concentration ($\mu\text{g}/\text{m}^3$)	Target Variable
DEWP/TEMP	Dew Point and Temperature ($^{\circ}\text{C}$)	Predictors
cbwd	Combined Wind Direction (NW, NE, SE, cv)	Segment
Iws	Cumulated wind speed (m/s)	Predictor
Ir/Is	Cumulated Rain/Snow hours	Precipitation Flag

4.4 Data Quality Issues

- **Nulls:** 2,067 rows (4.72%) missing in pm2.5.
- **Categorical Noise:** cbwd = "cv" (calm/variable) required specific handling as a valid meteorological state rather than missing data.

5 Cleaning and Transformation

5.1 Major Cleaning Steps

1. **Dropping Indices:** Removed the No column as it carried no analytical value.
2. **Datetime Synthesis:** Merged year, month, day, and hour into a single datetime64 column.
3. **Missing Value Imputation:** Applied forward-fill (then backward-fill) to pm2.5 nulls, assuming temporal autocorrelation.
4. **Snake case Normalisation:** Renamed columns (e.g., pm2.5 $\rightarrow$ pm2_5_cleaned).

5.2 Assumptions Made

- PM2.5 levels are temporally autocorrelated; hourly readings do not fluctuate wildly between adjacent hours.
- US EPA AQI breakpoints are valid proxies for health risk categorisation in this context.

5.3 Output Dataset Description

- **File:** beijing_pm25_cleaned.csv
- **Final Features:** 14 (Original 13 – No + datetime + season + aqi_category).

6 KPI Framework

6.1 Definitions and Formulae

KPI	Formula	Rationale
Annual Mean	$\bar{x}_{\text{PM}} = \frac{1}{n} \sum \text{PM}_{2.5}$	Primary compliance metric
Hazardous Rate	$\frac{\text{Days with Avg} > 150}{365} \times 100$	Quantifies acute risk frequency
WHO Ratio	$\frac{\bar{x}_{\text{PM}}}{15}$	Contextualises pollution against safety standards
AQI Distribution	$\frac{\text{Hours in Category}}{\text{Total Hours}}$	Visualises time-spent-at-risk

7 Exploratory Analysis

7.1 Major Trends

PM2.5 is heavily right-skewed (Mean: 97.8, Median: 72), driven by extreme spikes up to 994 $\mu\text{g}/\text{m}^3$. No consistent downward trend was observed between 2010 and 2014.

7.2 Segment-Level Insights

- **Diurnal:** Lowest readings occur at 2:00–4:00 PM (high mixing); peaks occur at night.
- **Wind Direction:** NW winds yield 69.8 $\mu\text{g}/\text{m}^3$ (lowest), while cv yields 124.5 $\mu\text{g}/\text{m}^3$ (highest).
- **Seasonality:** Winter (109.9) vs Summer (92.1) difference is statistically significant.

7.3 Visual Summaries

EDA yielded 11 plots including Monthly Average Time Series, Hourly Heatmaps, and AQI breakdown donut charts.

8 Statistical Analysis

8.1 Method Used

Due to the non-normal distribution of PM2.5 values, non-parametric statistical methods were applied to ensure reliable and assumption-free results.

- **Spearman Rank Correlation** was used to analyse relationships between PM2.5 and continuous variables such as wind speed and dew point.
- **Kruskal-Wallis Test** was used to compare pollution levels across categorical groupings such as seasons.
- **Mann-Whitney U Test** was used to evaluate the effect of rainfall on PM2.5 concentrations.

8.2 Results

Test	Stat / p-value	Business Finding
Spearman (Wind Speed)	$r = -0.35^{***}$	Strongest pollution reducer
Spearman (Dew Point)	$r = +0.30^{***}$	Humidity traps particulates
Mann-Whitney (Rain)	$p = 0.63$	Rain does not significantly clean air
Kruskal-Wallis (Season)	$H = 105.37^{***}$	Seasonal alerts are justified

***significant at $p < 0.001$

Wind speed shows a moderate negative correlation with PM2.5, confirming that higher wind speeds reduce pollution levels. Dew point carries a moderate positive correlation, indicating that elevated humidity increases particulate concentration. Seasonal differences are statistically significant, validating the use of season-based alert thresholds. Rainfall shows no significant impact on pollution at the hourly level.

8.3 Business Interpretation

- **Wind speed** is the most important natural factor for reducing pollution, making it the key variable for planning real-time interventions.
- **Seasonal variation** strongly supports the need for targeted winter pollution control strategies, particularly targeting coal-based heating.
- **High-humidity conditions** increase pollution risk and should be incorporated into forecasting and early-warning models.
- **Rainfall** is not a reliable mechanism for air purification at an hourly scale; emission-reduction policies must not depend on precipitation as a mitigating factor.

9 Dashboard Walkthrough

9.1 Dashboard Objective

The dashboard is designed to help decision-makers monitor pollution levels, identify high-risk periods, and understand the environmental factors that drive air quality deterioration. It supports both strategic planning and day-to-day operational response.

9.2 Executive View

The executive view provides high-level KPIs including Annual Mean PM2.5, WHO Ratio, and the number of hazardous days per year. It allows senior stakeholders to quickly assess overall air quality performance and year-on-year trends using the year filter.

9.3 Operational View

The operational view focuses on granular, drill-down analysis. It includes:

- **Hour × Month Heatmaps** to identify peak pollution periods throughout the day and across the calendar year.
- **Wind Direction Analysis** to understand dispersion patterns and identify natural “clean window” opportunities.

- **Seasonal Breakdowns** to highlight high-risk time-frames and support targeted intervention scheduling.

9.4 Filters and Interactivity

The dashboard includes interactive filters for:

- Year, Month, and Hour of Day
- AQI Category (Good, Moderate, Unhealthy, Hazardous)
- Season and Wind Direction

These filters allow users to drill into specific scenarios and isolate hazardous combinations. The dashboard follows a structured flow from *severity* → *timing* → *drivers*, enabling both high-level oversight and deep analytical investigation in a single interface.

10 Key Insights (Decision-Oriented)

The following insights are derived from the full analytical pipeline and are structured to directly inform policy and operational decisions:

1. PM2.5 levels consistently exceed WHO safe limits by 6–7 times every single year of the study period.
2. No significant improvement trend is observed from 2010 to 2014, indicating that existing measures were insufficient.
3. Approximately 21% of total recorded hours fall under hazardous pollution levels, meaning the air is unsafe for more than one in every five hours.
4. Approximately 1 in every 5 days per year is classified as hazardous, posing a recurring acute health risk.
5. Winter is the most polluted season, with mean levels 19% higher than summer, primarily driven by coal-based residential heating emissions.
6. Pollution levels peak during late-night and early-morning hours, coinciding with reduced atmospheric mixing and lower wind speeds.
7. Calm wind conditions ($cbwd = cv$) consistently produce the highest recorded PM2.5 levels, making stagnation the single most dangerous meteorological state.
8. Northwest winds significantly reduce PM2.5 concentrations, functioning as the city's primary natural air purification mechanism.
9. Wind speed is the strongest quantitative predictor of pollution reduction, with a Spearman correlation of $r = -0.35$ ($p < 0.001$).
10. Rainfall does not significantly reduce PM2.5 at the hourly level and cannot be relied upon as a natural mitigation mechanism.
11. Snow-related seasonal effects act as a confounding variable rather than a causal pollution driver and require careful treatment in modelling.

12. Summer months exhibit relatively lower pollution levels, offering a potential benchmark for evaluating the effectiveness of future emission-control programmes.

11 Recommendations

- **Recommendation 1:** Launch Winter-Priority Emission Reduction targeting coal heating (December–February).
- **Recommendation 2:** Deploy NW-Wind “Clean Window” alerts when NW winds exceed 21 m/s.
- **Recommendation 3:** Implement Calm-Wind Emergency Protocols (cbwd=cv) including industrial caps and traffic curtailment.
- **Recommendation 4:** Shift to Hourly AQI monitoring for health advisories to capture diurnal spikes.
- **Impact:** Potential 10–15% reduction in peak exposure and improved public health response timing.

12 Limitations and Next Steps

12.1 Limitations

- **Spatial:** Data represents a single monitoring station (US Embassy).
- **Policy:** Dataset predates major 2013 Action Plan interventions.
- **Causality:** No source apportionment; drivers are meteorological proxies only.

12.2 Future Work

- Multi-station analysis to map spatial pollution gradients across Beijing.
- Development of SARIMA or XGBoost models for predictive forecasting.
- Correlation of air quality data with hospital admission records.

Contribution Matrix

Name	ETL	EDA	Stats	Tableau	Report	PPT
Milind B.	Owner	Support	—	Support	Support	—
Hrishabh	—	Support	Support	Owner	—	—
Om	Support	—	—	—	Owner	Owner
Vishuti J.	—	Owner	Owner	—	Support	—
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Praveen N.	—	Support	Support	Owner	—	—